

Reminder and Notification Architecture (Design Definition)

Health Data Bank System

1. Purpose

This document defines the architecture of the Reminder and Notification system within the Health Data Bank. The objective is to establish a clear structure for how reminders are generated, processed, and converted into notifications, and how those notifications are delivered to users.

This document focuses on defining system components, responsibilities, and interactions. It does not describe implementation details.

2. Context

The reminder and notification functionality supports user engagement by prompting users to complete required actions such as form submissions or notifying them of important system updates.

Since the system is being developed in stages, this document serves as a design reference to ensure all components follow a consistent structure and integrate correctly.

3. Design Objectives

The system should:

- Support scheduled and condition-based reminders
- Convert reminder triggers into user-facing notifications
- Deliver notifications through defined channels
- Respect user preferences for notification delivery
- Maintain role-based access control (RBAC)
- Ensure privacy and security of user-specific information
- Record important system events for auditing
- Allow future expansion for additional notification types

4. Architecture Components

4.1 Reminder Rules Engine

Defines conditions under which reminders are triggered.

- Scheduled reminders (e.g., daily, weekly)
- Inactivity-based reminders (e.g., missing form submission)
- Threshold-based reminders (e.g., data exceeds defined limits)

Reminder rules are evaluated based on defined conditions such as missing form submissions, user inactivity over a specified period, or threshold-based checks on user data.

Reminder rules are defined using structured conditions that include:

- **Trigger condition** (e.g., missing form submission, inactivity duration exceeded, threshold violation)
- **Evaluation frequency** (e.g., daily, weekly, real-time checks)
- **Target users or roles** (e.g., patient, provider, researcher)
- **Cooldown or repetition control** to prevent duplicate or excessive reminders

Each rule is evaluated by the system during scheduled execution cycles.

When a rule condition is satisfied, a reminder event is generated for the affected user.

The system ensures:

- Rules are applied consistently across all users
- Duplicate reminders are avoided within defined time windows
- Rules can be extended in the future to support additional conditions without affecting existing functionality

4.2 Reminder Trigger Process

A system process that evaluates reminder rules at regular intervals.

- Runs periodically
- Identifies users meeting reminder conditions

4.3 Reminder Event Generator

Generates structured reminder events when conditions are satisfied.

Each event includes:

- User identifier
- Reminder type
- Message content
- Timestamp

4.4 Notification Structure

Defines the format of notifications.

Each notification includes:

- User identifier
- Message content
- Notification type (reminder, alert)
- Status (read/unread)
- Timestamp

4.5 Notification Storage

Stores notifications in a structured format.

- Enables retrieval for dashboard display
- Maintains notification history
- Supports read/unread tracking

4.6 Notification Preferences

Defines how user preferences affect notification delivery.

- System checks user preferences before sending notifications
- Preferred channels are used when available
- Default settings are applied if preferences are not configured

4.7 Notification Delivery Service

Responsible for delivering notifications to users.

- Supports in-app notifications (dashboard/notification center)
- Supports email notifications
- Handles delivery attempts and failures
- May retry or switch channels when delivery fails

4.8 Audit Logging

Defines how system events are recorded.

The system should log:

- Reminder generation events
- Notification creation events
- Notification delivery status (success/failure)
- Delivery failures and retry attempts
- User interaction (e.g., notification read status, where applicable)

This ensures traceability and supports debugging and monitoring.

5. Architecture Interaction Flow

Reminder events are generated by the Reminder Rules Engine and passed to the Notification Service, which creates, stores, and prepares notifications for delivery through the selected channels.

The Notification Delivery Service is responsible for routing notifications through the appropriate channels and handling delivery outcomes.

6. Reminder and Notification Processing Flow

The system behavior is as follows:

- The system evaluates reminder conditions at regular intervals
- Users with pending actions are identified
- RBAC and privacy checks are applied
- Reminder events are generated
- Reminder events are logged
- Notifications are created
- Notifications are stored
- User preferences are checked
- Delivery channel is selected based on user preferences
- Notifications are delivered through selected channels
- Delivery success or failure is recorded
- Notifications remain available until user interaction

7. Example Scenario

If a user has not submitted required health data:

- The system detects the missing action
- A reminder condition is satisfied
- A reminder event is generated and logged
- A notification is created
- The system checks user preferences
- The notification is delivered through the selected channel
- Delivery status is recorded
- The user views the notification and it is marked as read

8. Role-Based Access Control (RBAC)

The system must enforce:

- Users only receive notifications relevant to their own data
- Notifications are not accessible across unauthorized roles
- All access follows defined RBAC policies

This ensures privacy and secure data handling.

9. Design Constraints

The system must:

- Integrate with existing modules such as form submission and dashboards
- Maintain privacy and security standards
- Support scalability and future enhancements
- Remain independent of implementation details

10. Conclusion

This document defines the architecture of the Reminder and Notification system by outlining its components, responsibilities, and interaction flow. It ensures consistency across system components and provides a structured foundation for future development.

Reminder and Notification Workflow

